

Solutions to MP352 Problem Set 5

①

P. 9

(a) The "stone" is a unit of force, not mass - "weight" is specifically the force that a mass exerts on a scale under the influence of a constant gravitational field, so $11\frac{1}{2}$ stone is an amount of force. $1 \text{ stone} = 14 \text{ pounds}$, because a pound is also a unit of force; thus, $11\frac{1}{2} \text{ stone}$ is 161 lb . Now, we often say that $1 \text{ kg} = 2.2 \text{ lb}$, but this is actually not correct, as a kilogramme is a unit of mass. What this conversion actually says is that a mass of one kilogramme, in a constant gravitational field of 9.81 m.s^{-2} , exerts a force of 2.2 pounds on the earth. Thus, 2.2 pounds is actually 9.81 newtons , not 1 kg .

However, this is not important, because we can still convert between pounds and kilograms: $W = 2.2 \text{ lb} \Rightarrow M = 1 \text{ kg}$, so $W = 161 \text{ lb} \Rightarrow M = \frac{161}{2.2} \cdot 1 \text{ kg} = 73.2 \text{ kg}$. Thus, a man weighing $11\frac{1}{2} \text{ stone}$ has a mass of 73.2 kg .

To convert to TeV/c^2 we use the following: 1 kg is equivalent to a rest energy of $Mc^2 = 1 \text{ kg} \cdot (3 \times 10^8 \text{ m.s}^{-1})^2 = 9 \times 10^{16} \text{ J}$. But $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$, so

$$1 \text{ kg} = 9 \times 10^{16} \text{ J}/c^2$$

$$= \frac{9 \times 10^{16}}{1.6 \times 10^{-19}} \text{ eV}/c^2 = 5.63 \times 10^{35} \text{ eV}/c^2$$

$$1 \text{ TeV} = 10^{12} \text{ eV}, \text{ so } 1 \text{ kg} = 5.63 \times 10^{23} \text{ TeV}/c^2. \text{ Thus,}$$

$$11\frac{1}{2} \text{ stone} \Rightarrow 73.2 \text{ kg} = 73.2 (5.63 \times 10^{23} \text{ TeV}/c^2) = \boxed{4.12 \times 10^{25} \text{ TeV}/c^2}$$

(b) In (a), we already showed that 1 kg corresponds to a rest energy of $9 \times 10^{16} \text{ J}$. Thus,

$$20 \text{ g} = 0.02 \text{ kg}$$

corresponds to a rest energy of

$$0.02 \cdot 9 \times 10^{16} \text{ J} = \boxed{1.8 \times 10^{15} \text{ J}}$$

②

(c) The momentum of this football is $p = (0.42 \text{ kg})(5 \text{ m.s}^{-1})$

$$= 2.1 \text{ kg.m.s}^{-1}, \text{ Thus,}$$

$$pc = (2.1 \text{ kg.m.s}^{-1})(3 \times 10^8 \text{ m.s}^{-1}) = 6.3 \times 10^8 \text{ J.}$$

We know $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$ and $1 \text{ MeV} = 10^6 \text{ eV}$, so

$$pc = \frac{6.3 \times 10^8}{1.6 \times 10^{-19}} \cdot 10^{-6} \text{ MeV} = 3.93 \times 10^{21} \text{ MeV.}$$

Dividing by c gives the result we want: $p = 3.93 \times 10^{21} \text{ MeV/c}$

All of the above are supposed to illustrate the suitability of different unit systems: for human-sized quantities, TeV, MeV, etc. are a bit unwieldy. However, for subatomic particles, they're far more practical than kg, kg.m.s⁻¹, J, etc.

P. 2

This is easiest if it's all done in the centre-of-mass (CM) frame of the electron and positron. Let's say, in this frame, the electron momentum is

$\vec{p}$. Because it's the CM frame, $\vec{P}_{\text{tot}} = 0$, so the positron has momentum $-\vec{p}$. Therefore, the two particles have the same energy $E_e = \sqrt{\vec{p}^2 c^2 + m_e^2 c^4}$

(because they have the same mass). Thus, the total CM energy is $2\sqrt{\vec{p}^2 c^2 + m_e^2 c^4}$.

Now, if these can annihilate to a single photon, momentum conservation says the photon must have zero momentum in this frame ($\vec{P}_{\text{out}} = \vec{P}_{\text{in}}$) but a total energy of $2\sqrt{\vec{p}^2 c^2 + m_e^2 c^4}$ ($E_{\text{out}} = E_{\text{in}}$).

But a photon is massless, so it has to obey the mass-shell condition $E = |\vec{p}|c$, but it can't have zero momentum and

nonzero energy, (the smallest $2\sqrt{\vec{p}^2 c^2 + m_e^2 c^4}$ can be is $2m_e c^2$, when both the electron and positron are at rest).

Thus, $e^+ e^- \rightarrow \gamma$ is not possible (at least not if the photon is physical).

Put another way, $P_{\text{in}}^2 = p_{\text{e}^+}^2 + p_{\text{e}^-}^2$, so $-P_{\text{in}} \cdot P_{\text{in}} = m_e^2 c^2 + 2p_{\text{e}^+} \cdot p_{\text{e}^-} + m_e^2 c^2$

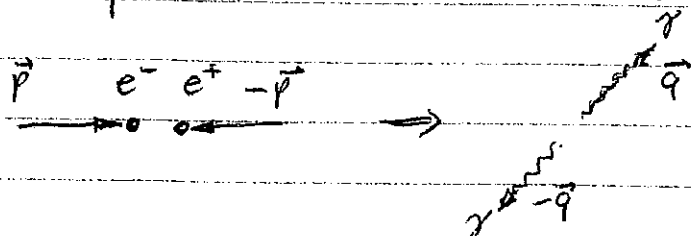
$$= 2m_e^2 c^2 + 2p_{\text{e}^+} \cdot p_{\text{e}^-} = 2m_e^2 c^2 + 2(E_{\text{e}^+} E_{\text{e}^-} - \vec{p}_{\text{e}^+} \cdot \vec{p}_{\text{e}^-}). \text{ But } E_{\text{e}^+} E_{\text{e}^-} - \vec{p}_{\text{e}^+} \cdot \vec{p}_{\text{e}^-} \geq 0$$

in all frames, so $-P_{\text{in}} \cdot P_{\text{in}} \geq 2m_e^2 c^2$. But $P_{\text{out}} \cdot P_{\text{out}} = p_{\gamma} \cdot p_{\gamma} = 0$, so

again we see a contradiction. (In fact, this is the general argument saying that two particles cannot combine into a single photon if either is massive.)

(8)

Now let's look at $e^+ + e^- \rightarrow \gamma + \gamma$. Let $\vec{q}$ be the momentum of one of the photons; then the other must be $-\vec{q}$ in order to have $\vec{P}_{out} = \vec{P}_{in} = \vec{0}$.



The total final energy is

Therefore

$$|\vec{q}|c + |-\vec{q}|c = 2|\vec{q}|c,$$

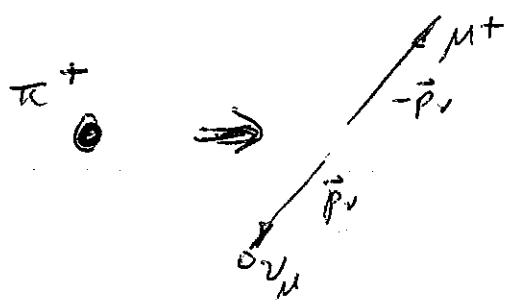
but it's perfectly okay

for this to be $2\sqrt{|\vec{p}|^2c^2 + m_e^2c^4}$; in fact, we see that each final photon has the exact same energy as the initial electron (or positron) did.

Thus, $e^+ + e^- \rightarrow \gamma + \gamma$ is possible.

P. 3

(a) Let's do momentum and energy conservation separately, first: in the pion's rest frame, $\vec{P}_{in} = \text{total initial 3-momentum} = \vec{0}$. Since $\vec{P}_{out} = \text{total final momentum} = \vec{P}_{in}$, this means $\vec{P}_{out} = \vec{P}_\mu + \vec{P}_\nu = \vec{0}$,



ie. $\vec{P}_\mu = -\vec{P}_\nu$. Let $|\vec{p}|$ be the magnitude of both output momenta ($|\vec{p}| = |\vec{p}_\nu| = |\vec{p}_\mu|$).

That's momentum conservation taken care of: now we do the same for energy, $E_{in} = E_{out} \Rightarrow E_\pi = E_\mu + E_\nu$. Since the pion is at rest, its energy is just its rest energy $E_\pi = m_\pi c^2$.

$E_\mu = \sqrt{|\vec{p}|^2c^2 + m_\mu^2c^4}$ (since $|\vec{p}_\mu| = |\vec{p}|$), and since the neutrino is massless, $E_\nu = |\vec{p}|c$. Therefore,

$$m_\pi c^2 = \sqrt{|\vec{p}|^2c^2 + m_\mu^2c^4} + |\vec{p}|c$$

Subtracting $|\vec{p}|c$ and squaring both sides gives

$$(m_\pi c^2 - |\vec{p}|c)^2 = |\vec{p}|^2c^2 + m_\mu^2c^4 \Rightarrow m_\pi^2c^4 - 2m_\pi|\vec{p}|c^3 + |\vec{p}|^2c^2 = |\vec{p}|^2c^2 + m_\mu^2c^4.$$

$|\vec{p}|^2c^2$ cancels out from both sides, and simple algebra gives the answer:

$$|\vec{p}| = \frac{m_\pi^2c^2 - m_\mu^2c^2}{2m_\pi c}.$$

(5)

(b) Remember that the speed of a relativistic particle is $\frac{|\vec{p}|c}{E}$; since $E_\nu = |\vec{p}|c$ for the neutrino, this gives $|\vec{v}_\nu| = c$, as expected. For the muon, we see

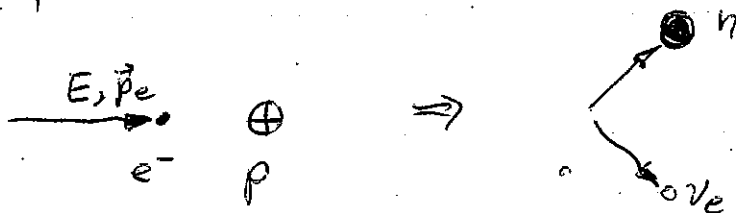
$$\begin{aligned} E_\mu &= \sqrt{|\vec{p}_\mu|^2 c^2 + m_\mu^2 c^4} = \sqrt{\left(\frac{m_\pi^2 - m_\mu^2}{2m_\pi}\right)^2 c^2 + m_\mu^2 c^4} \\ &= \sqrt{\frac{(m_\pi^4 - 2m_\pi^2 m_\mu^2 + m_\mu^4) + 4m_\pi^2 m_\mu^2}{4m_\pi^2}} c^2 \\ &= \frac{m_\pi^2 + m_\mu^2}{2m_\pi} c^2 \quad (= m_\pi c^2 - |\vec{p}|c \text{ as well}) \end{aligned}$$

so

$$|\vec{v}_\mu| = \frac{m_\pi^2 - m_\mu^2}{m_\pi^2 + m_\mu^2} c = \frac{(139.57)^2 - (105.66)^2}{(139.57)^2 + (105.66)^2} c = \boxed{0.27c}$$

or about $8.1 \times 10^7 \text{ m.s}^{-1}$.

P.4



In the proton's rest frame, the total initial energy is $E_{in} = E + m_p c^2$ and the total momentum is $\vec{P}_{in} = \vec{p}_e$. The total centre-of-mass energy of the reaction is then E_{cm} , where

$$E_{cm}^2/c^2 = -\vec{P}_{in} \cdot \vec{P}_{in} = E_{in}^2/c^2 - |\vec{P}_{in}|^2 = (E + m_p c^2)^2/c^2 - |\vec{p}_e|^2$$

$$= E_e^2/c^2 + 2Em_p + m_p^2 c^2 - |\vec{p}_e|^2 = 2Em_p + (m_p^2 + m_e^2)c^2$$

because $E^2 = p_e^2 c^2 + m_e^2 c^4$.

Now, $-\vec{P}_{in} \cdot \vec{P}_{in}$ has the same value in any frame, and since $\vec{P}_{in}^M = \vec{P}_{out}^M$, $-\vec{P}_{out} \cdot \vec{P}_{out} = 2Em_p + (m_p^2 + m_e^2)c^2$ in any frame as well. This includes the C.M. frame of the $n\nu_e$ final state: in this frame, the neutron and neutrino have momenta of equal magnitude (call it q) but opposite direction, i.e. $\vec{P}_{out} = 0$. But E_{out}

$$= E_n + E_\nu = \sqrt{q^2 c^2 + m_n^2 c^4} + q c, \text{ so}$$

$$-\vec{P}_{out} \cdot \vec{P}_{out} = E_{out}^2/c^2 - |\vec{P}_{out}|^2 = (\sqrt{q^2 c^2 + m_n^2 c^4} + q c)^2/c^2$$

$$= 2Em_p + (m_p^2 + m_e^2)c^2$$

(5)

Thus, if $\vec{q}$ is the momentum of the neutrino in the final CM frame, then the initial electron energy in the proton rest frame is

$$E = \frac{1}{2m_p} \left| \frac{(\sqrt{q^2 c^2 + m_n^2 c^4} + q c)^2}{c^2} - (m_p^2 + m_e^2) c^2 \right|$$

(a) The threshold energy for $e^- + p \rightarrow n + \nu e$ is the smallest energy E such that the reaction can take place. We see above that E is related to q , the magnitude of the neutrino in the final CM frame; thus, if we control the value of q which minimizes E , then we've got our threshold energy. It's pretty obvious that as $q \rightarrow 0$, E gets smaller and smaller. Now, q can never be exactly zero - a neutrino with zero momentum also has zero energy - but q can in principle be arbitrarily close to zero. Thus the absolute minimum value of E is

$$E_{\text{threshold}} = \lim_{q \rightarrow 0} \frac{1}{2m_p} \left| \frac{(\sqrt{q^2 c^2 + m_n^2 c^4} + q c)^2}{c^2} - (m_p^2 + m_e^2) c^2 \right|$$

$$= \boxed{\frac{m_n^2 - m_p^2 - m_e^2}{2m_p} c^2}$$

or, putting in the masses, $\boxed{E_{\text{threshold}} = 1.3 \text{ MeV}}$.

(b) Now, if we think of the interior of a star as a gas of electrons at temperature T , then the kinetic energy of each electron is $\frac{3}{2} k_B T$, giving a total energy of $m_e c^2 + \frac{3}{2} k_B T$. So the minimum temperature at which the $e^- + p \rightarrow n + \nu e$ reaction can start is that at which $m_e c^2 + \frac{3}{2} k_B T = E_{\text{threshold}}$.

$$\text{Thus } T = \frac{E_{\text{threshold}} - m_e c^2}{\frac{3}{2} k_B} = \frac{2}{3} \frac{1.3 \text{ MeV} - 0.51 \text{ MeV}}{8.62 \times 10^{-5} \text{ eV} \cdot \text{K}^{-1}}$$

$$= \boxed{6.11 \times 10^9 \text{ K}}$$

or a bit over six billion kelvin. More accurate calculations lower this a bit to 5 billion K, but the essential idea is the same: There has to be a minimum electron energy to even initiate the reaction.